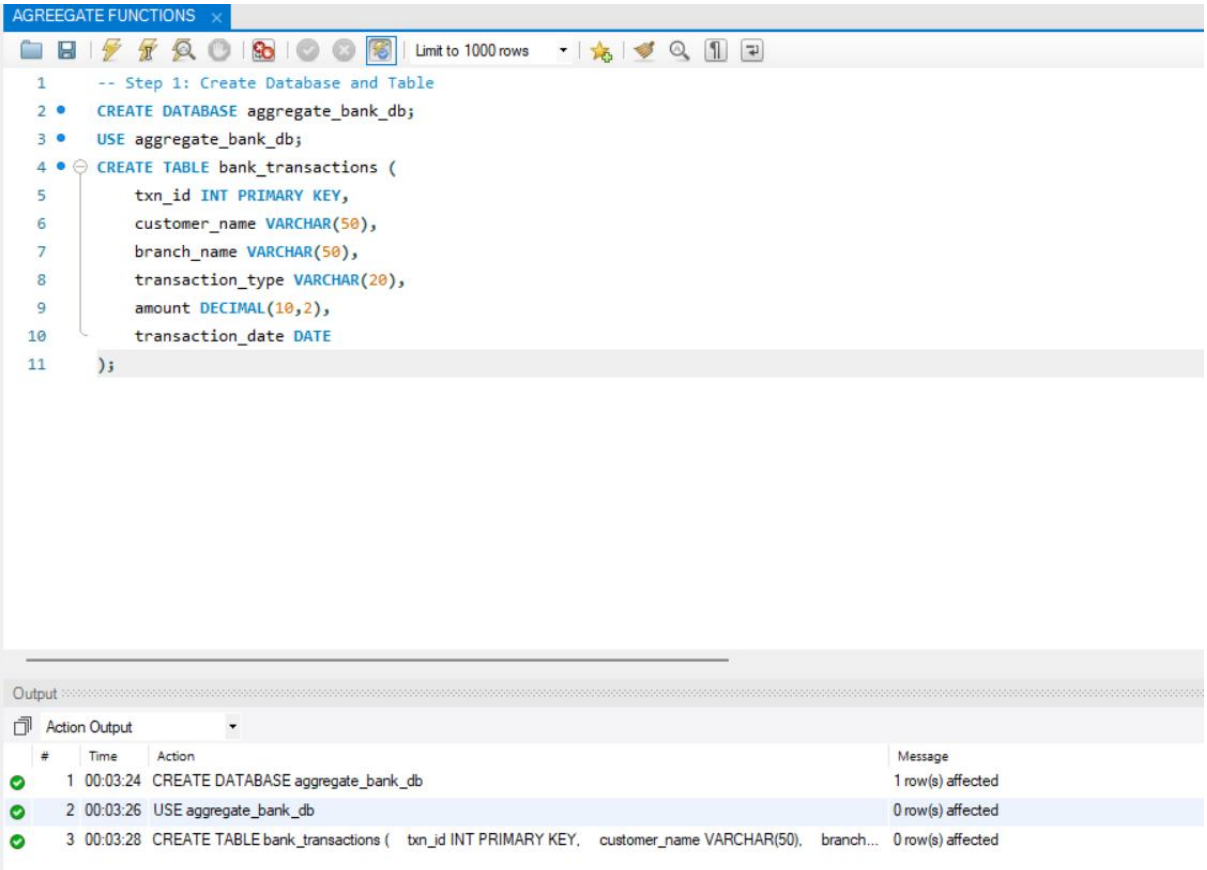
AGREEGATE FUNCTIONS

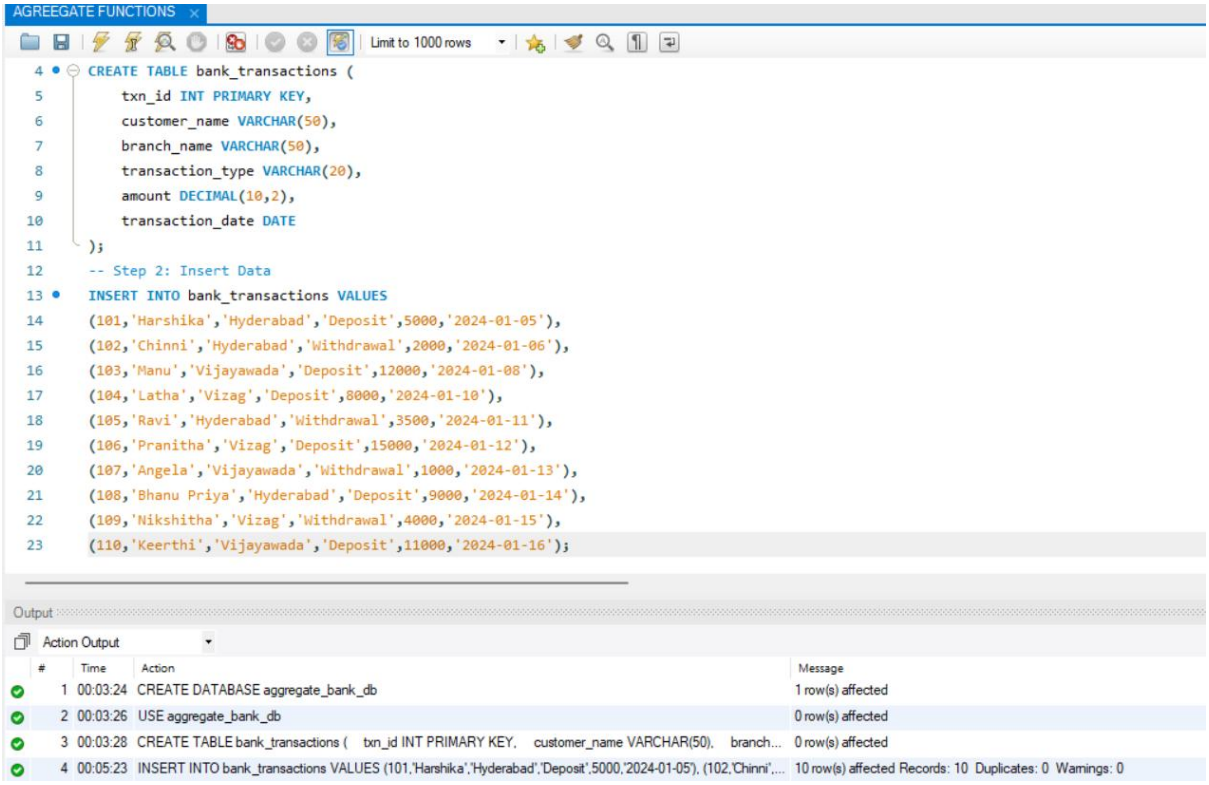
Ili.Harshika (2520030607)

Question :

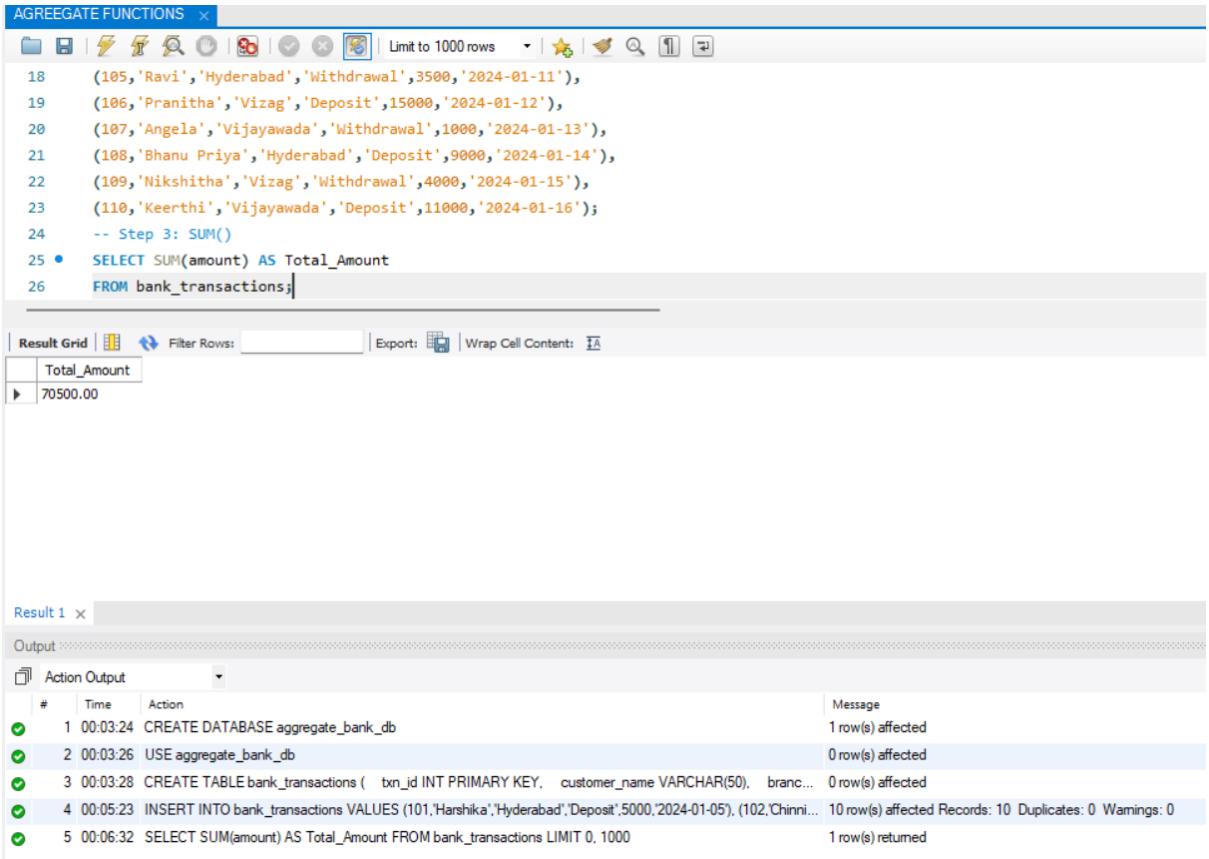
Perform SQL aggregate operations on bank transaction data using SUM(), AVG(), MAX(), MIN(), and COUNT() along with WHERE, GROUP BY, HAVING, and ORDER BY clauses to analyse transaction amounts, branches, deposits, withdrawals, and transaction counts.

Step 1 : Create Database and Table





Step 2 : AGREEGATE FUNCTIONS



AGREEGATE FUNCTIONS

```
    (108,'Bhanu Priya','Hyderabad','Deposit',9000,'2024-01-14'),
    (109,'Nikshitha','Vizag','Withdrawal',4000,'2024-01-15'),
    (110,'Keerthi','Vijayawada','Deposit',11000,'2024-01-16');
    -- Step 3: SUM()
    SELECT SUM(amount) AS Total_Amount
    FROM bank_transactions;
    -- Step 4: AVG()
    SELECT AVG(amount) AS Average_Transaction
    FROM bank_transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Average_Transaction
7050.000000

Result 2

Output

Action Output

#	Time	Action	Message
1	00:03:24	CREATE DATABASE aggregate_bank_db	1 row(s) affected
2	00:03:26	USE aggregate_bank_db	0 row(s) affected
3	00:03:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	0 row(s) affected
4	00:05:23	INSERT INTO bank_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Chinni...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	00:06:32	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	00:07:11	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

AGREEGATE FUNCTIONS

```
    -- Step 3: SUM()
    SELECT SUM(amount) AS Total_Amount
    FROM bank_transactions;
    -- Step 4: AVG()
    SELECT AVG(amount) AS Average_Transaction
    FROM bank_transactions;
    -- Step 5: MAX()
    SELECT MAX(amount) AS Highest_Transaction
    FROM bank_transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Highest_Transaction
15000.00

Result 3

Output

Action Output

#	Time	Action	Message
2	00:03:26	USE aggregate_bank_db	0 row(s) affected
3	00:03:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected
4	00:05:23	INSERT INTO bank_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Chi...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	00:06:32	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	00:07:11	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	00:07:47	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

AGREEGATE FUNCTIONS

```
27      -- Step 4: AVG()
28      SELECT AVG(amount) AS Average_Transaction
29      FROM bank_transactions;
30      -- Step 5: MAX()
31      SELECT MAX(amount) AS Highest_Transaction
32      FROM bank_transactions;
33      -- Step 6: MIN()
34      SELECT MIN(amount) AS Lowest_Transaction
35      FROM bank_transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Lowest_Transaction
1000.00

Result 4

Output

Action Output

#	Time	Action	Message
3	00:03:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected
4	00:05:23	INSERT INTO bank_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Chi...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	00:06:32	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	00:07:11	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	00:07:47	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	00:08:17	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

AGREEGATE FUNCTIONS

```
30      -- Step 5: MAX()
31      SELECT MAX(amount) AS Highest_Transaction
32      FROM bank_transactions;
33      -- Step 6: MIN()
34      SELECT MIN(amount) AS Lowest_Transaction
35      FROM bank_transactions;
36      -- Step 7: COUNT()
37      SELECT COUNT(*) AS Total_Transactions
38      FROM bank_transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Transactions
10

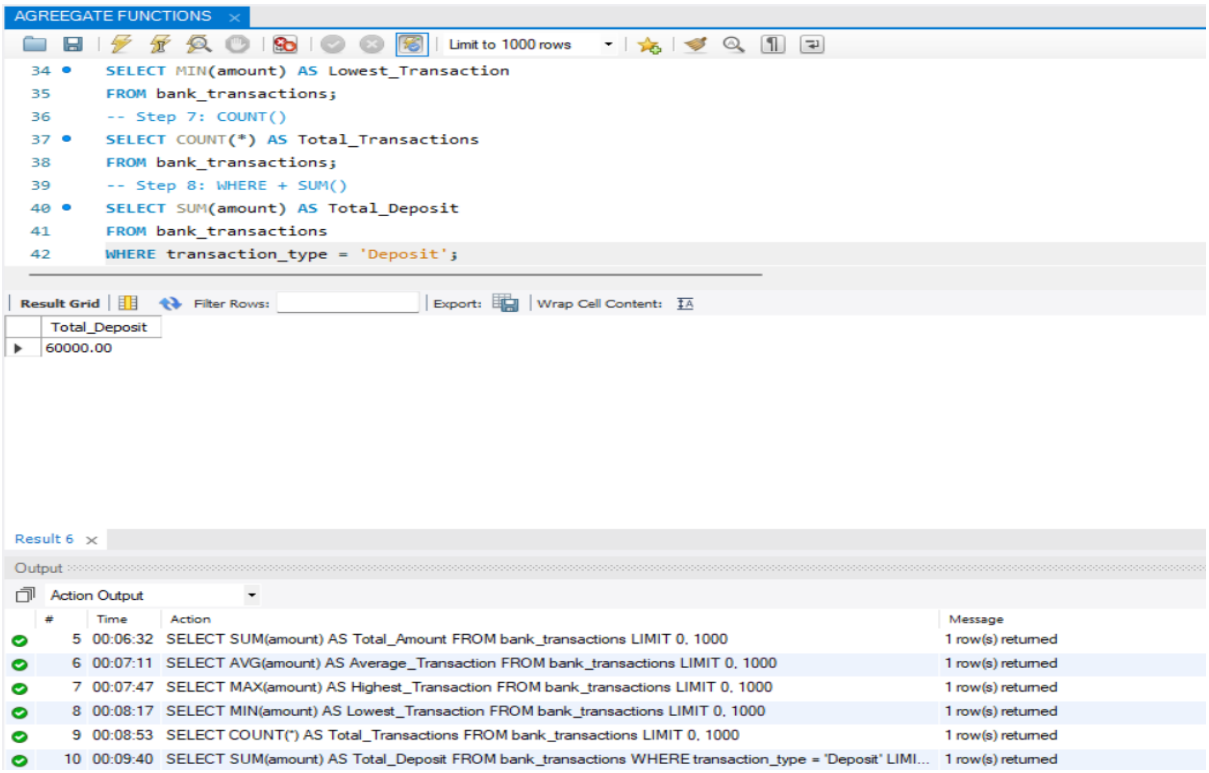
Result 5

Output

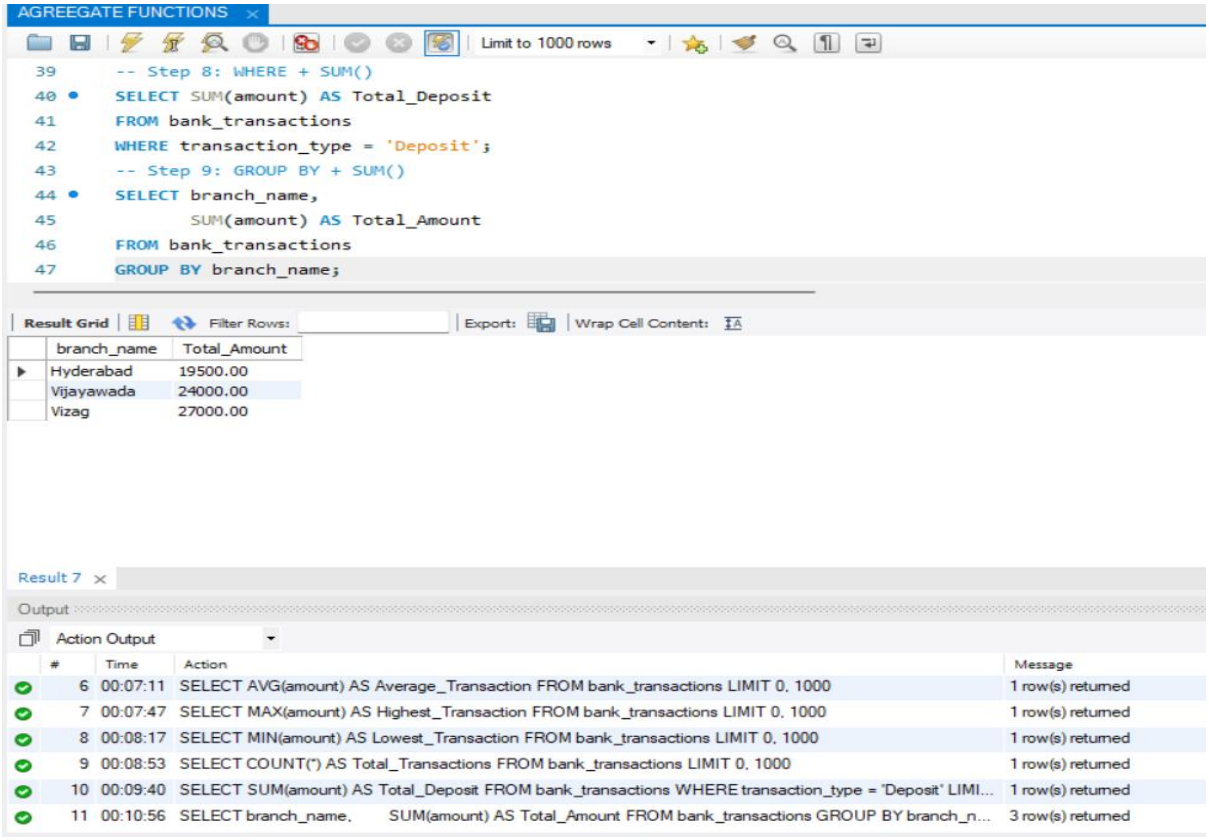
Action Output

#	Time	Action	Message
4	00:05:23	INSERT INTO bank_transactions VALUES (101,'Harshika','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Chi...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	00:06:32	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	00:07:11	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	00:07:47	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	00:08:17	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	00:08:53	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

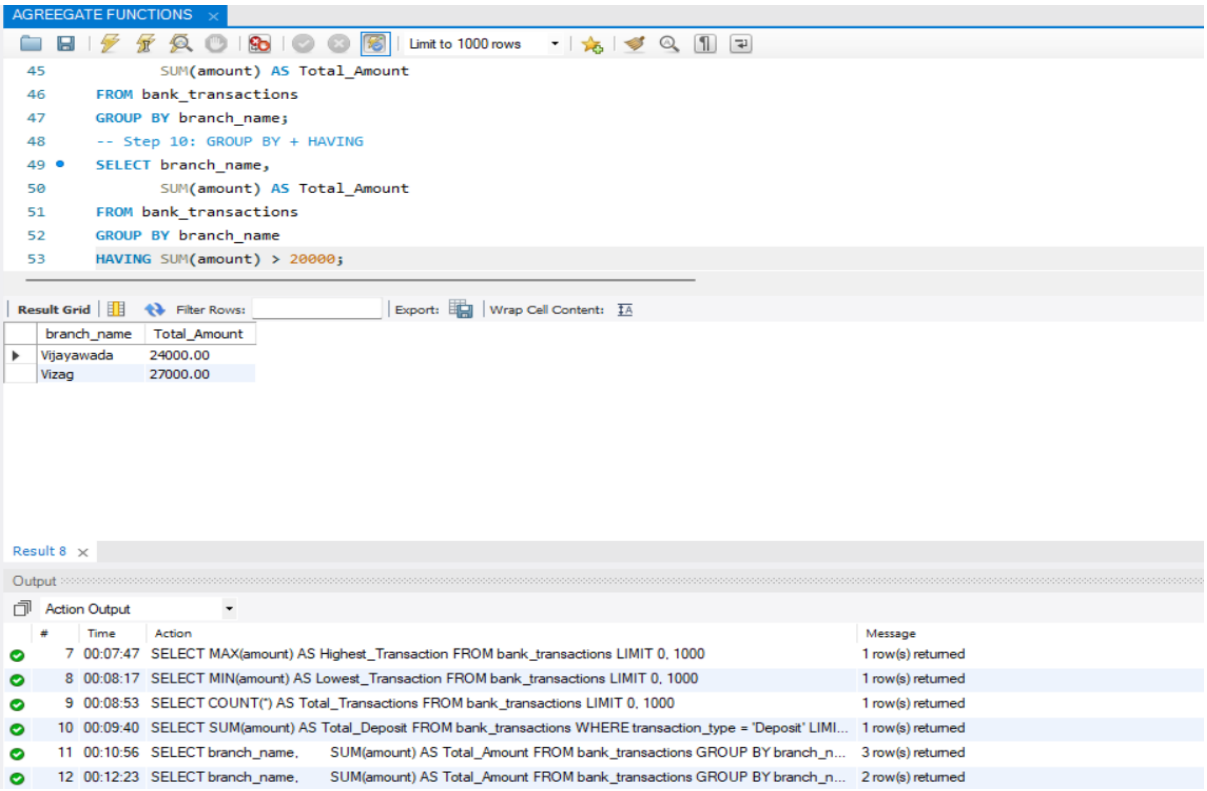
Step 3 : WHERE + AGGREGATE



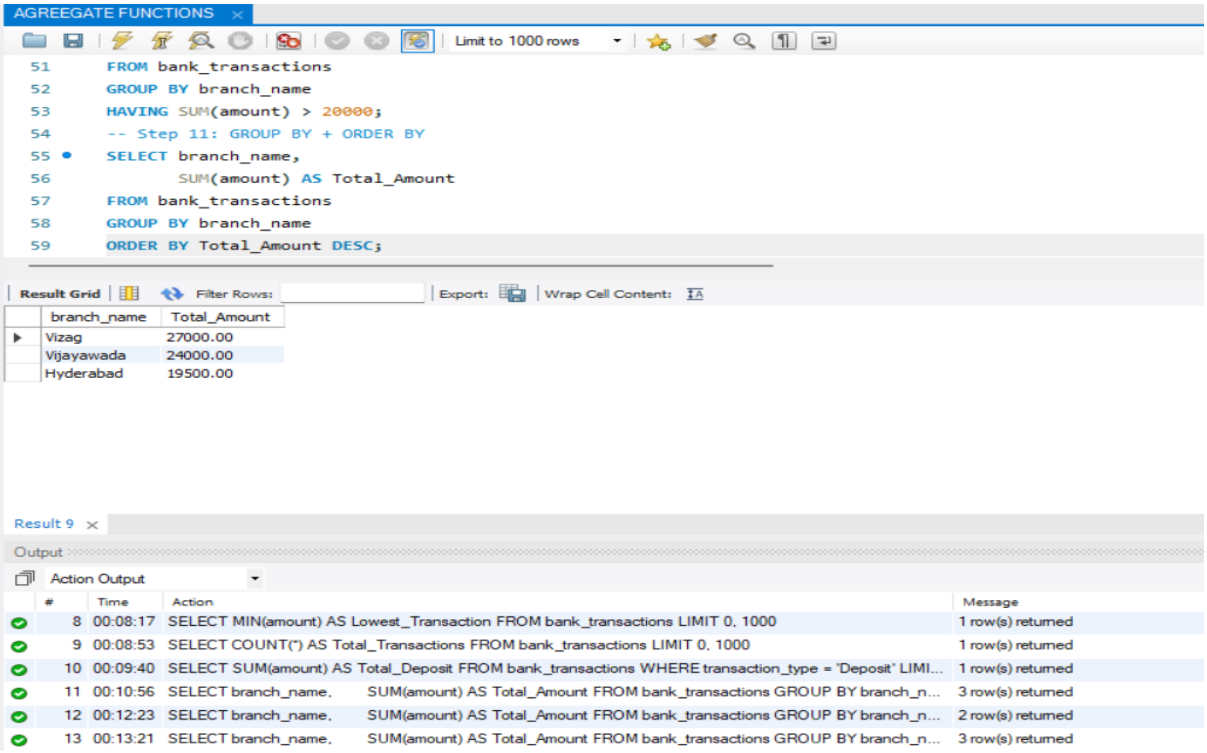
Step 4 : GROUP BY



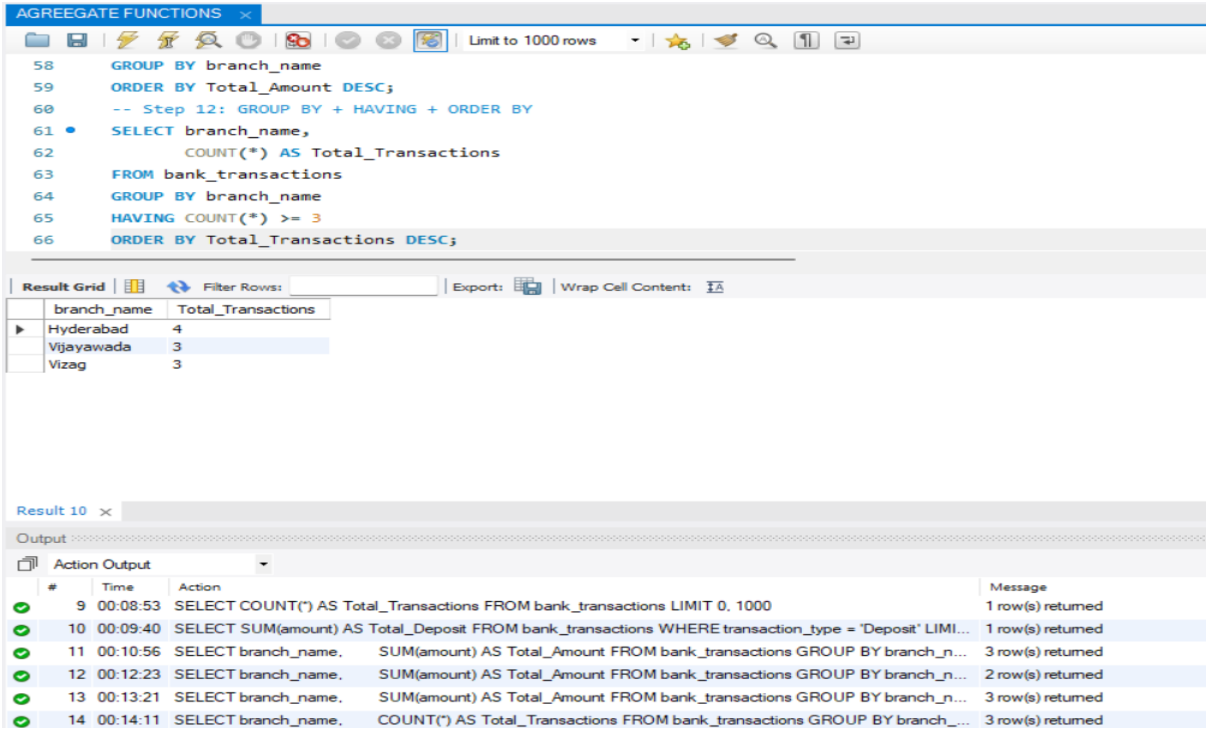
Step 5 : GROUP BY + HAVING



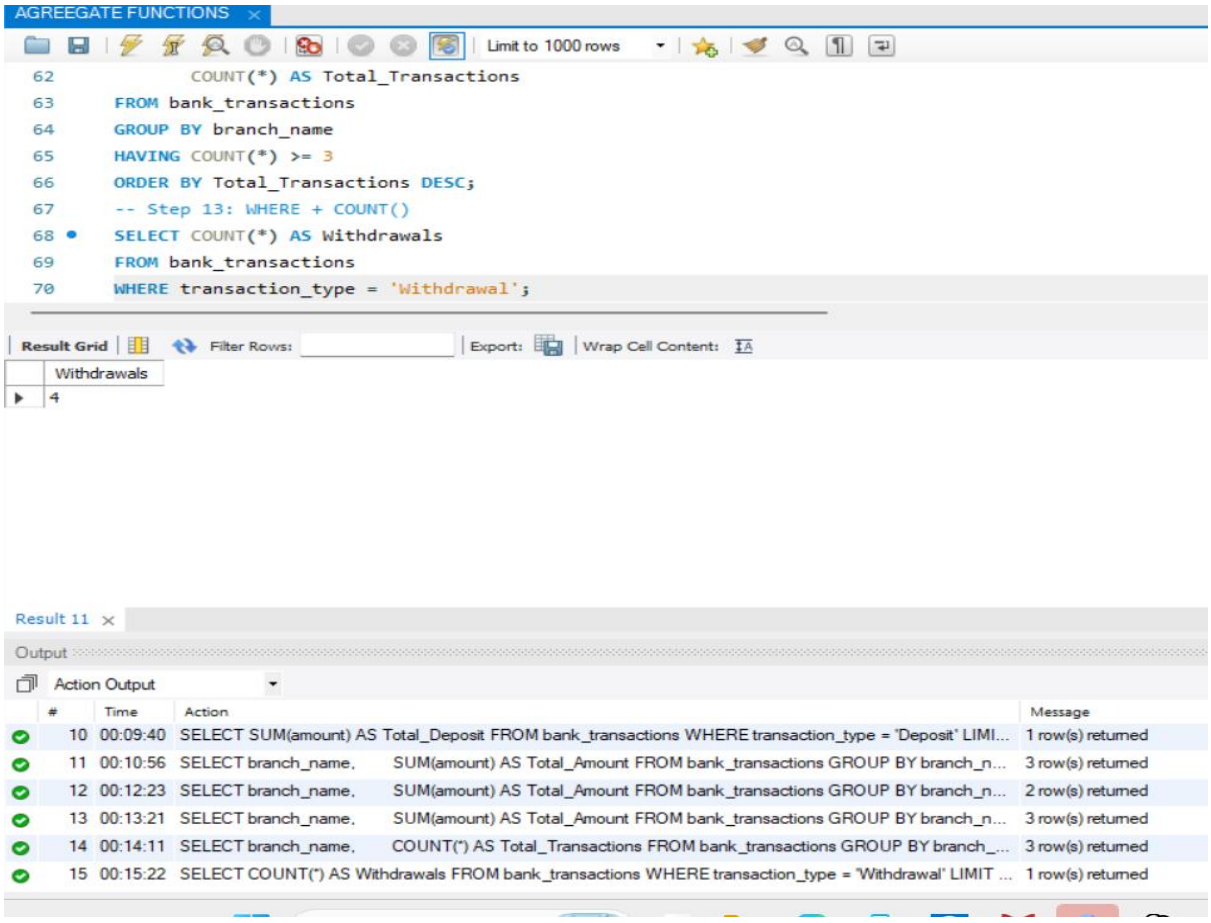
Step 6 : GROUP BY + ORDER BY



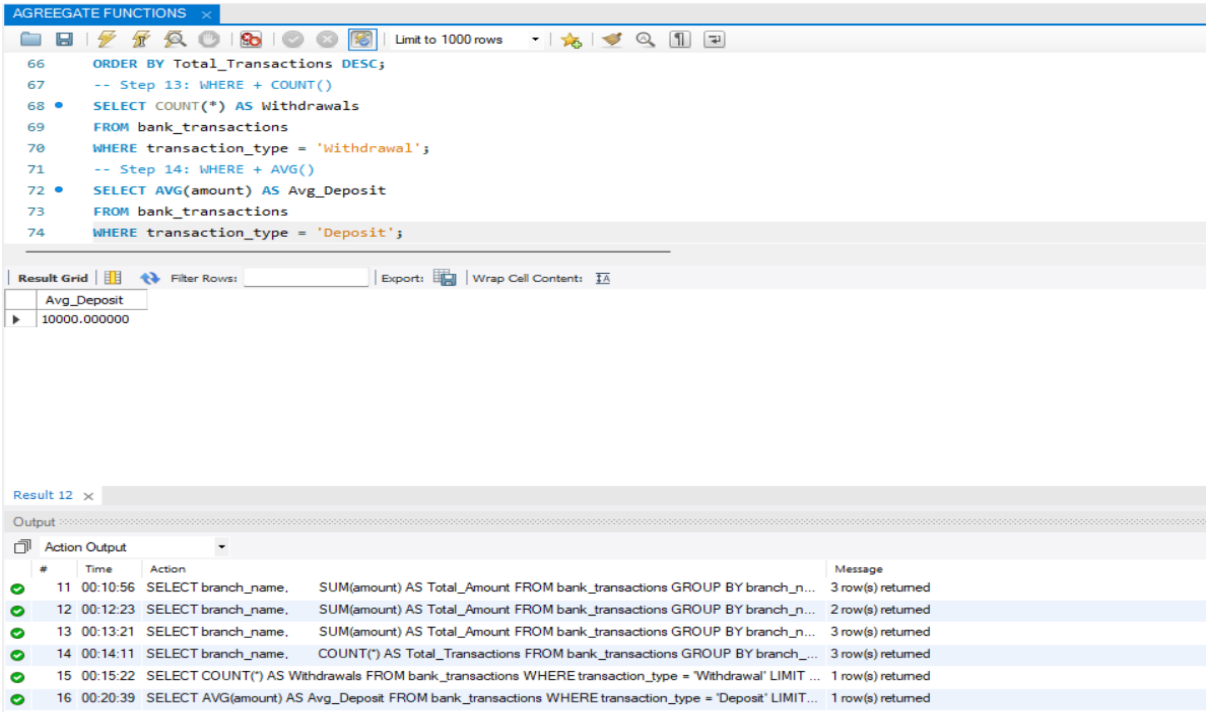
Step 7: GROUP BY + HAVING + ORDER BY



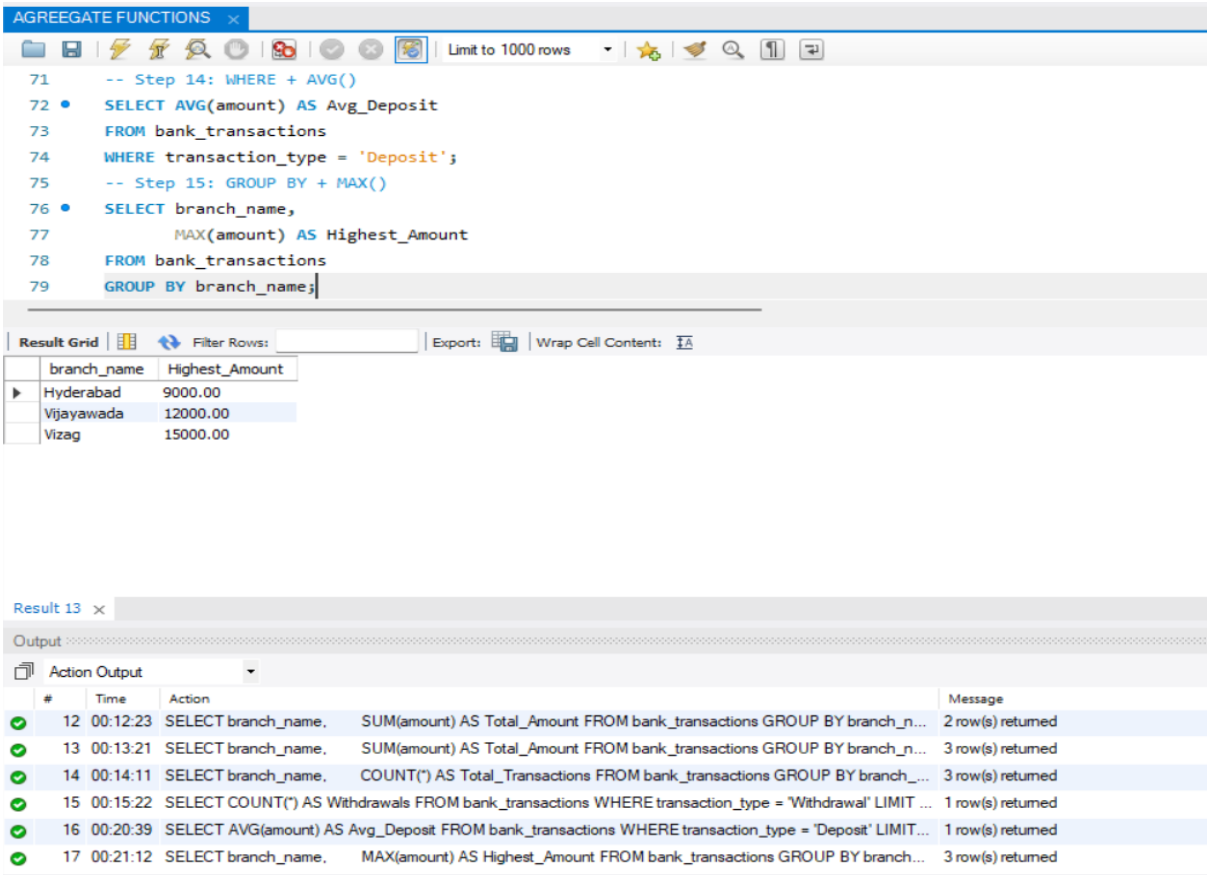
Step 8 : WHERE + COUNT()



Step 9 : WHERE + AVG()



Step 10 : GROUP BY + MAX()



Step 11: GROUP BY + MIN()

AGREEGATE FUNCTIONS

```
76 •  SELECT branch_name,
77            MAX(amount) AS Highest_Amount
78    FROM bank_transactions
79    GROUP BY branch_name;
80    -- Step 16: GROUP BY + MIN()
81 •  SELECT branch_name,
82            MIN(amount) AS Lowest_Amount
83    FROM bank_transactions
84    GROUP BY branch_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

Result 14

Output

Action Output

#	Time	Action	Message
13	00:13:21	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
14	00:14:11	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
15	00:15:22	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
16	00:20:39	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned
17	00:21:12	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned
18	00:21:47	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned

Step 12: WHERE + ORDER BY

AGREEGATE FUNCTIONS

```
81 •  SELECT branch_name,
82            MIN(amount) AS Lowest_Amount
83    FROM bank_transactions
84    GROUP BY branch_name;
85    -- Step 17: WHERE + ORDER BY
86 •  SELECT *
87    FROM bank_transactions
88    WHERE amount > 8000
89    ORDER BY amount DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Pranitha	Vizag	Deposit	15000.00	2024-01-12
103	Manu	Vijayawada	Deposit	12000.00	2024-01-08
110	Keerthi	Vijayawada	Deposit	11000.00	2024-01-16
108	Bhanu Priya	Hyderabad	Deposit	9000.00	2024-01-14
NULL	NULL	NULL	NULL	NULL	NULL

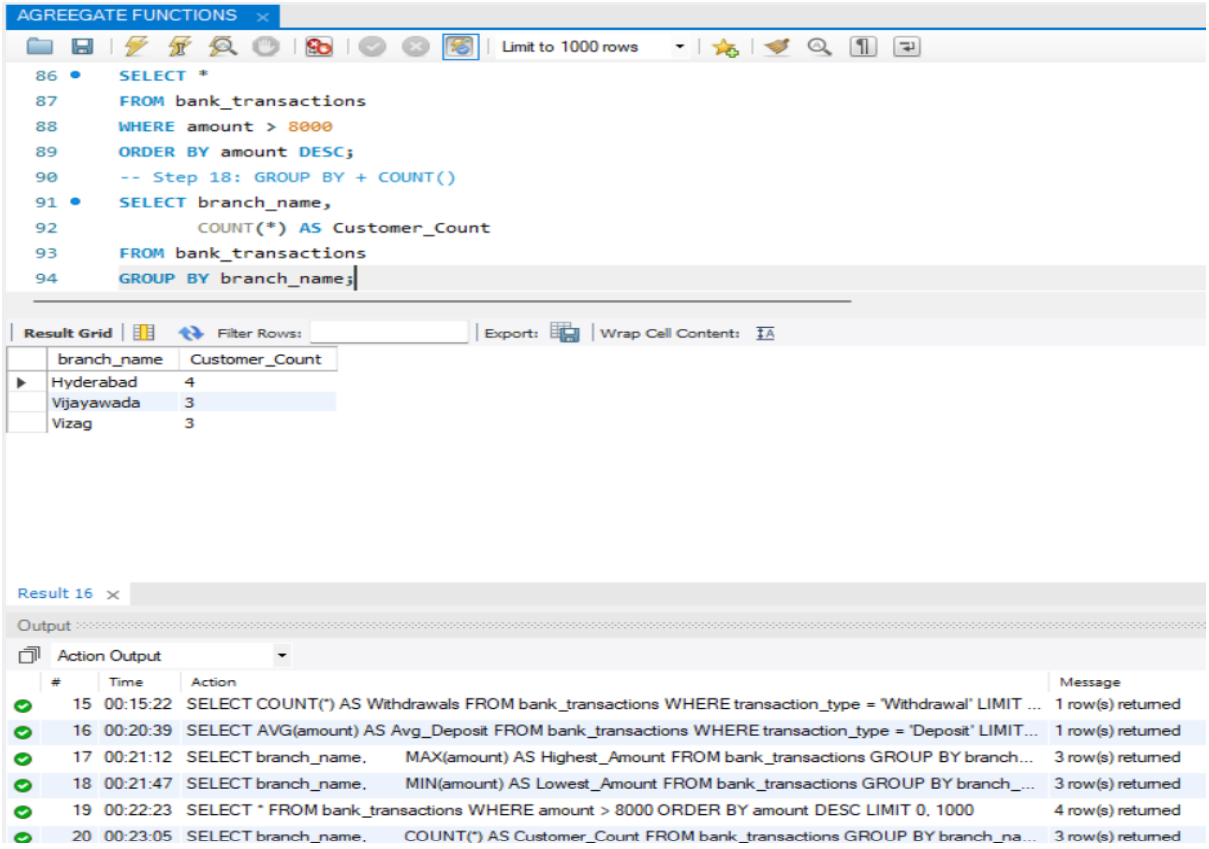
bank_transactions 15

Output

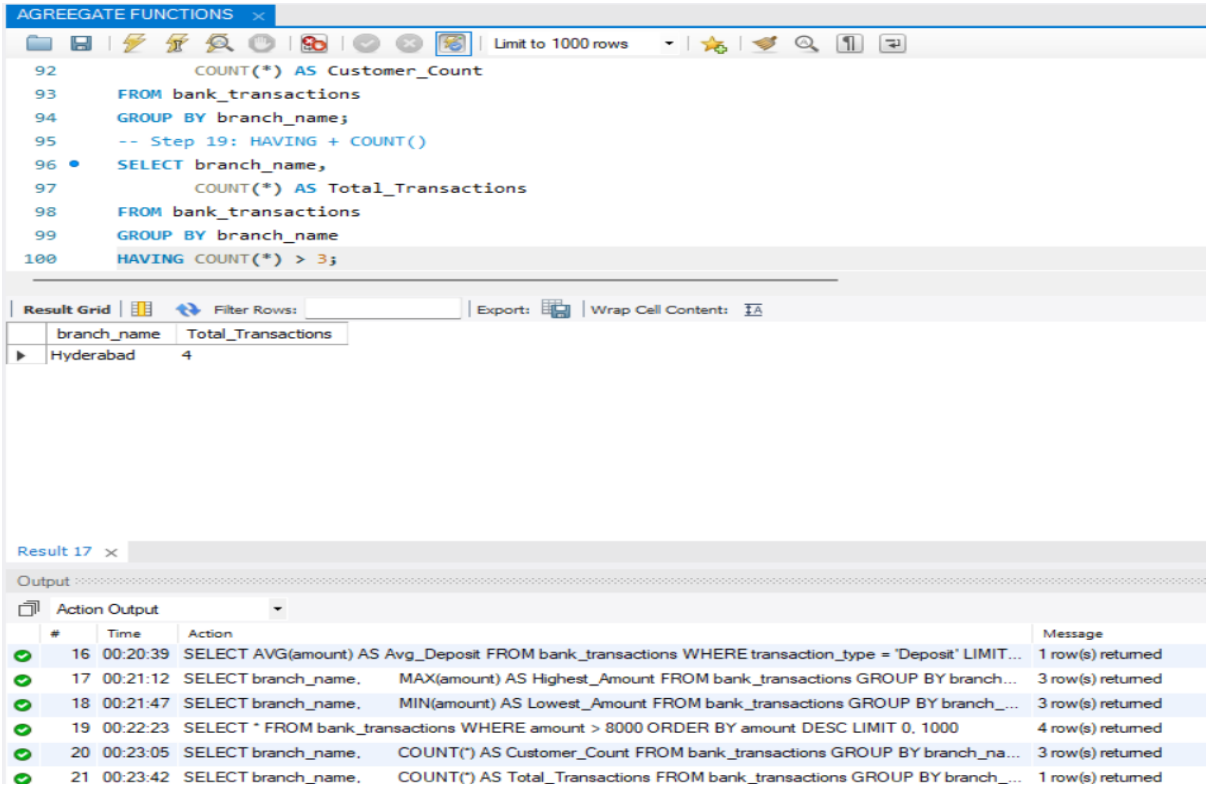
Action Output

#	Time	Action	Message
14	00:14:11	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
15	00:15:22	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal' LIMIT ...	1 row(s) returned
16	00:20:39	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT...	1 row(s) returned
17	00:21:12	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch...	3 row(s) returned
18	00:21:47	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned
19	00:22:23	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned

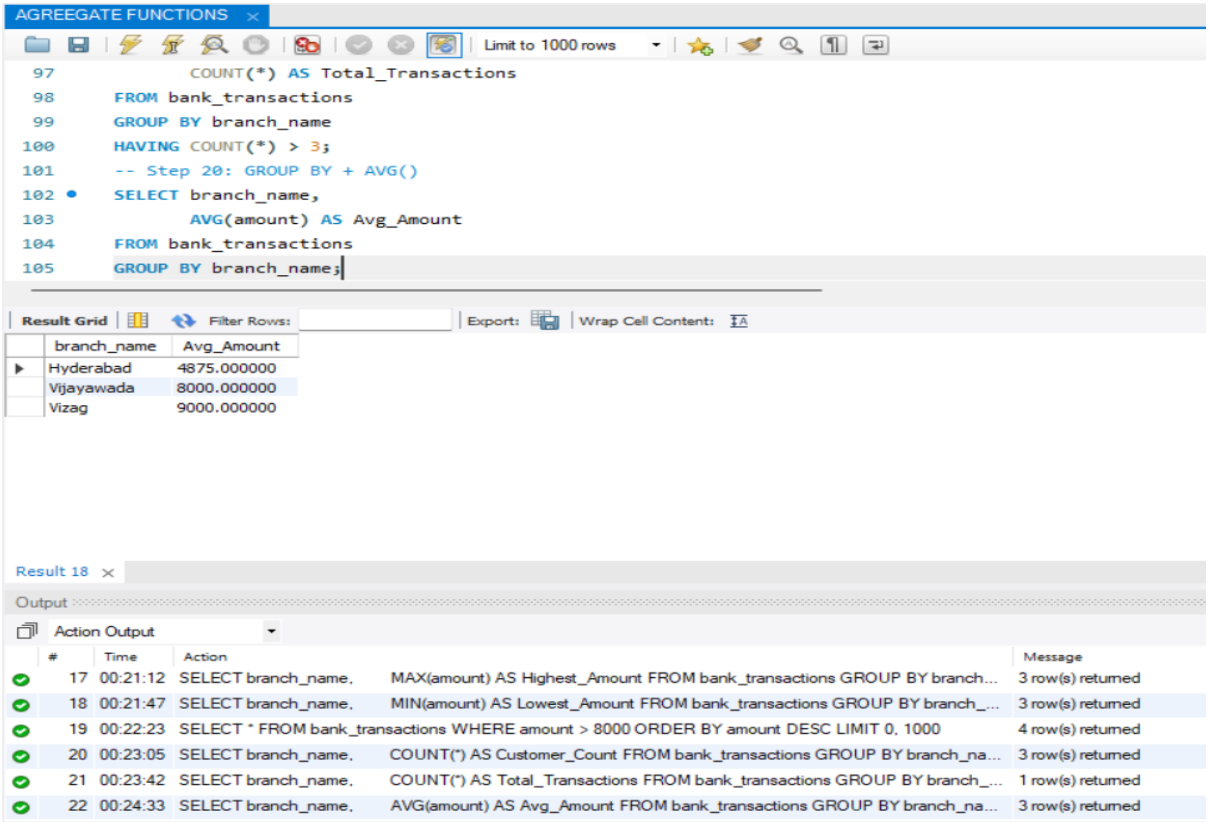
Step 13: GROUP BY + COUNT()



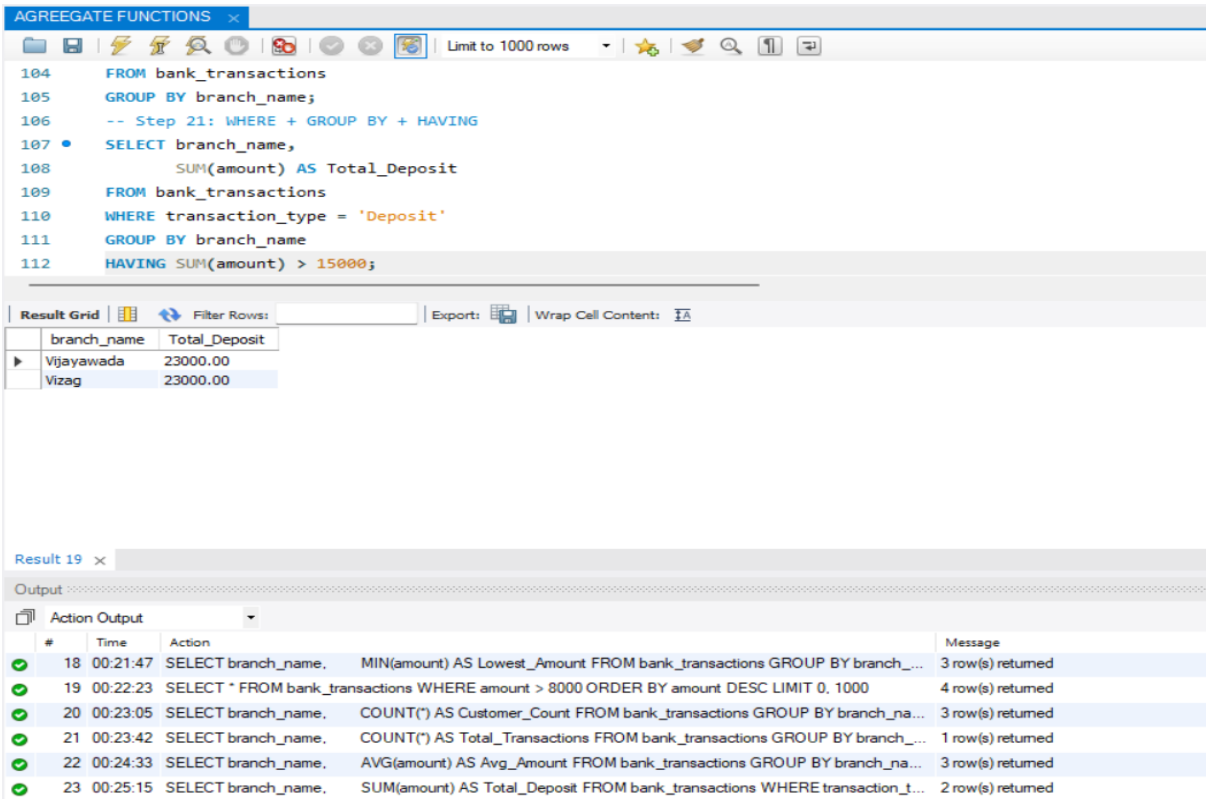
Step 14: HAVING + COUNT()



Step 15: GROUP BY + AVG()



Step 16: WHERE + GROUP BY + HAVING



Step 17: GROUP BY + HAVING + ORDER BY

